44.1: Osmoregulation balances the uptake and loss of water and solutes

Osmoregulation requires controlled movement of water and solutes across plasma membrane. Ultimately, the driving force for the movement of both water and solutes is a concentration gradient of one or more solutes across the membrane.

Osmosis and Osmolarity

Water enters and leaves cells by osmosis, which occurs when two solutions separated by a membrane differ in total solute concentration. The unit of measurement for solute concentration is **osmolarity**, the number of moles of solute per liter of solution. The osmolarity of human blood is about 300milliosmoles per liter (mOsm/L), whereas that of seawater is about 1,000 mOsm/L.

Two solutions with the same osmolarity are said to be *isoosmotic*. If a selectively permeable membrane separates the solutions, water molecules will continually cross the membrane at equal rates in both directions. Thus, there is no *net* movement of water by osmosis between isoosmotic solutions. When two solutions differ in osmolarity, the solution with the higher concentration of solutes is said to be *hyperosmotic*, and the more dilute solution is said to be *hypoosmotic*. Water flows by osmosis from a hypoosmotic solution to a hyperosmotic one, thus reducing the concentration difference for both solutes and free water (see the diagram on the first page of the chapter).

In this chapter, we use the terms *isoosmotic, hypoosmotic*, and *hyperosmotic*, which refer specifically to osmolarity, instead of *isotonic, hypotonic*, and *hypertonic*. The latter set of terms applies to the response of animal cells—whether they swell or shrink—in solutions of known solute concentrations.

Osmoregulatory Challenges and Mechanisms

An animal can maintain water balance in two ways. One is to be an **osmoconformer**: to be isoosmotic with its surroundings. All osmoconformers are marine animals. Because an osmoconformer's internal osmolarity is the same as that of its environment, there is no tendency to gain or lose water. Many osmoconformers live in water that has a stable composition and hence have a constant internal osmolarity.

The second way to maintain water balance is to be an **osmoregulator**: to control internal osmolarity independent of that of the external environment. In a hypoosmotic environment, an osmoregulator must discharge excess water. In a hyperosmotic environment, it must instead take in water to offset osmotic loss.

Osmoregulation enables animals to live in environments that are uninhabitable for osmoconformers, such as freshwater and terrestrial habitats, or to move between marine and freshwater environments **(Figure 44.2)**.

Figure 44.2

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    Sockeye salmon (*Oncorhynchus nerka*), osmoregulators that
migrate between rivers and the ocean.
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Sockeye salmon swim through a river.

Whether osmoconformers or osmoregulators, most animals cannot tolerate substantial changes in external osmolarity and are said to be *stenohaline* (from the Greek *stenos*, narrow, and *halos*, salt). In contrast, *euryhaline* animals (from the Greek *eurys*, broad) can survive large fluctuations in external osmolarity. Euryhaline osmoconformers include barnacles and mussels in estuaries that are alternately exposed to fresh and salt water; euryhaline osmoregulators include striped bass and the various species of salmon (see Figure 44.2).

Next, we'll examine adaptations for osmoregulation that have evolved in marine, freshwater, and terrestrial animals.

Marine Animals

Most marine invertebrates are osmoconformers. Their osmolarity is the same as that of seawater. Therefore, they face no substantial challenges in water balance. Nevertheless, they actively transport *specific* solutes that they maintain at levels different from those in the ocean. For example, homeostatic mechanisms in the Atlantic lobster (*Homarus americanus*) maintain a magnesium ion (Mg^{2+}) concentration in hemolymph (circulatory fluid) of less than 9 mM (millimolar, or 10^{-3} mol/L), far below the 50 mM concentration of Mg^{2+} in their environment.

Two osmoregulatory strategies evolved among marine vertebrates that address the challenges of a strongly dehydrating environment. One is found among marine "bony fishes," a group that includes ray-finned and lobe-finned fishes. The other is found in marine sharks and most other chondrichthyans (cartilaginous animals; see Concept 34.3).

Cod, shown in **Figure 44.3a**, and other marine bony fishes constantly lose water by osmosis. They balance water loss by drinking a lot of seawater. The excess salts ingested with seawater are eliminated through the gills and kidneys.

Figure 44.3

Osmoregulation in marine and freshwater bony fishes: a comparison.

Osmoregulation in marine and freshwater bony fishes: a comparison.

Part a. Osmoregulation in a marine fish (saltwater). Fish in saltwater gains water and salt ions from food (by mouth) and gains water and salt ions from drinking seawater. Excretion results in the loss of salt ions from gills, as well as the loss of salt ions and small amounts of water in scanty urine from the kidneys. Water is also lost via osmosis through the gills and other parts of the body surface.

Part b. Osmoregulation in a freshwater fish. In fresh water, the fish gains water and some ions in food (by mouth). There is uptake of salt ions by the gills, as well as osmotic water gain through gills and other parts of the body surface. Excretion results in the loss of salt ions and large amounts of water in dilute urine from the kidneys.

Like bony fishes, sharks have an internal salt concentration much lower than that of seawater. However, shark tissues have a high concentration of several other solutes, including urea and another organic molecule, trimethylamine oxide (TMAO). Because the total solute concentration is somewhat higher than 1,000 mOsm/L, water slowly *enters* the shark's body by osmosis and in food (sharks do not drink).

The small influx of water into the shark's body is disposed of in urine produced by the kidneys. The urine also removes some of the salt that diffuses into the shark's body; the rest is lost in feces or is secreted from a specialized gland.

Freshwater Animals

The osmoregulatory problems of freshwater animals are the opposite of those of marine animals. The body fluids of freshwater animals must be hyperosmotic because animal cells cannot tolerate salt concentrations as low as that of lake or river water. Since freshwater animals have internal fluids with an osmolarity higher than that of their surroundings, they face the problem of gaining water by osmosis. For the perch and many other freshwater animals, water balance relies on excreting large amounts of very dilute urine and drinking almost no water **(Figure 44.3b)**. In addition, salts lost by diffusion and in the urine are replenished by eating and by taking up salt across their gills.

Salmon and other euryhaline fishes that migrate between fresh water and seawater undergo dramatic changes in osmoregulatory status. When living in rivers and streams, salmon osmoregulate like other freshwater fishes, producing large amounts of dilute urine and taking up salt from the dilute environment through their gills. When they migrate to the ocean, salmon acclimatize (see Concept 40.2). They produce more of the steroid hormone cortisol, which increases the number and size of specialized salt-secreting cells.

These and other physiological changes enable salmon in salt water to excrete excess salt from their gills and produce only small amounts of urine—just like bony fishes that spend their entire lives in salt water.

Animals That Live in Temporary Waters

Extreme dehydration, or *desiccation*, is fatal for most animals. However, a few aquatic invertebrates that live in temporary ponds and in films of water around soil particles can lose almost all their body water and survive. These animals enter a dormant state when their habitats dry up, an adaptation called **anhydrobiosis** ("life without water"). Among the most striking examples are the tardigrades, or water bears, tiny invertebrates less than 1 mm long **(Figure 44.4)**. In their active, hydrated state, they contain about 85% water by weight, but they can dehydrate to less than 2% water and survive in an inactive state, dry as dust, for a decade or more. Just add water, and within hours the rehydrated tardigrades are moving about and feeding.

Figure 44.4

Anhydrobiosis.

A hydrated tardigrade, on a 50 micrometer scale, is at least twice the size of a dehydrated tardigrade.

When hydrated, tardigrades (SEM images) inhabit temporary ponds as well as droplets of water in soil and on moist plants.

Anhydrobiosis requires adaptations that keep cell membranes intact. Researchers are just beginning to learn how tardigrades survive drying out, but studies of anhydrobiotic roundworms (phylum Nematoda; see Concept 33.4) show that desiccated individuals contain large amounts of sugars. In particular, a disaccharide called trehalose seems to protect the cells by replacing the water that is normally associated with proteins and membrane lipids. Many insects that survive freezing in the winter also use trehalose as a membrane protectant, as do some plants resistant to desiccation.

Recently, scientists began applying lessons learned from the study of anhydrobiosis to the preservation of biological materials. Traditionally, samples of protein, DNA, and cells have been kept in ultracold freezers −80°C, consuming large amounts of energy. Today, the manufacture of materials modeled after the protectants of anhydrobiotic species has enabled room temperature storage of such samples.

Land Animals

The threat of dehydration is a major regulatory problem for terrestrial plants and animals. Adaptations that reduce water loss are key to survival on land. Much as a waxy cuticle contributes to the success of land

plants, the body coverings of most terrestrial animals help prevent dehydration. Examples are the waxy layers of insect exoskeletons, the shells of land snails, and the layers of dead, keratinized skin cells covering most terrestrial vertebrates, including humans. Many terrestrial animals, especially desert-dwellers, are nocturnal, which reduces evaporative water loss because of the lower temperature and higher humidity of night air.

Despite anatomical and behavioral adaptations that conserve water, most terrestrial animals lose water through a variety of routes: in urine and feces, across the skin, and from the epithelial surfaces of gas exchange organs and airways. Land animals maintain water balance by drinking and eating moist foods and by producing water metabolically through cellular respiration.

A number of desert animals are well enough adapted for minimizing water loss that they can survive for long periods of time without drinking. Camels, for example, tolerate a 7°C rise in body temperature, greatly reducing the amount of water lost in sweat production. They can also lose 25% of their body water and survive. (In contrast, a human who loses half this fraction of body water will die from heart failure.) In the **Scientific Skills Exercise**, you can examine water balance in another desert species, the sandy inland mouse.

Scientific Skills Exercise

Describing and Interpreting Quantitative Data

AP Science Practices 2, 4, 5

How Do Desert Mice Maintain Osmotic Homeostasis? The sandy inland mouse (now known scientifically as *Pseudomys hermannsburgensis*) is an Australian desert mammal that can survive indefinitely on a diet of dried seeds without drinking water. To study this species' adaptations to its arid environment, researchers conducted a laboratory experiment in which they controlled access to water. In this exercise, you will analyze some of the data from the experiment.

A mouse.

How the Experiment Was Done Nine captured mice were kept in an environmentally controlled room and given birdseed (10% water by weight) to eat. In part A of the study, the mice had unlimited access to tap water for drinking; in part B of the study, the mice were not given any drinking water for 35 days, similar to conditions in their natural habitat. At the end of parts A and B, the researchers measured the osmolarity and urea concentration of the urine and blood of each mouse. The mice were also weighed three times a week.

Data from the Experiment

A table illustrates the access to water for mice in the part A and B and mean osmolarity and mean urea concentration.

Access to Water	Mean Osmolarity (milliosmoles per liter)	Mean Urea Concentration (millimolar)
Urine	Blood	Urine
Part A: Unlimited	490	350
Part B: None	4,700	320
In part A, the mice drank about 33 percent of their body weight each day. The change in body weight during the study was negligible for all mice.		

Data from R. E. MacMillen et al., Water economy and energy metabolism of the sandy inland mouse, *Leggadina hermannsburgensis, Journal of Mammalogy* 53:529–539 (1972).

INTERPRET THE DATA

- In words, describe how the data differ between the unlimited water and no-water conditions for the following:
- osmolarity of urine,
- osmolarity of blood,
- urea concentration in urine,
- urea concentration in blood.
- Does this data set provide evidence of homeostatic regulation? Explain.
-
    - Calculate the ratio of urine osmolarity to blood osmolarity for mice with unlimited access to water.
- Calculate this ratio for mice with no access to water.
- What conclusion can you draw from these ratios?

- If you learned that the amount of urine produced in part A were different from that in part B, how would that affect your calculation? Explain.

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

Energetics of Osmoregulation

Maintaining an osmolarity difference between an animal's body and its external environment carries an energy cost. Because diffusion tends to equalize concentrations in a system, osmoregulators must expend energy to maintain the osmotic gradients that cause water to move in or out. They do so by using active transport to manipulate solute concentrations in their body fluids.

The energy cost of osmoregulation depends on how different an animal's osmolarity is from its surroundings, how easily water and solutes can move across the animal's surface, and how much work is required to pump solutes across the membrane. Osmoregulation accounts for 5% or more of the resting metabolic rate of many fishes. For brine shrimp, small crustaceans that live in extremely salty lakes, the gradient between internal and external osmolarity is very large, and the cost of osmoregulation is correspondingly high—as much as 30% of the resting metabolic rate.

The energy cost to an animal of maintaining water and salt balance is minimized by having body fluids that are adapted to the salinity of the animal's habitat. Thus, the body fluids of most animals that live in fresh water (which has an osmolarity of 0.5–15 mOsm/L) have lower solute concentrations than the body fluids of their closest relatives that live in seawater (1,000 mOsm/L). For instance, whereas marine molluscs have body fluids with solute concentrations of approximately 1,000 mOsm/L, some freshwater molluscs maintain the osmolarity of their body fluids at just 40 mOsm/L. In each case, minimizing the osmotic difference between body fluids and the surrounding environment decreases the energy cost of osmoregulation.

Transport Epithelia in Osmoregulation

Although the ultimate function of osmoregulation is to control solute concentrations in cells, most animals do this indirectly, managing instead the solute content of the body fluid that bathes the cells. In insects and other animals with an open circulatory system, the fluid surrounding cells is hemolymph. In vertebrates and other animals with a closed circulatory system, the cells are bathed in an interstitial fluid that contains a mixture of solutes controlled indirectly by the blood. Maintaining the composition of such fluids depends on structures ranging from individual cells that regulate solute movement to complex organs such as the vertebrate kidney.

In most animals, osmoregulation and metabolic waste disposal rely on **transport epithelia**—one or more layers of epithelial cells specialized for moving particular solutes in controlled amounts in specific directions. Transport epithelia are typically arranged into tubular networks with extensive surface areas. Some transport epithelia face the outside environment directly, whereas others line channels connected to the outside by an opening on the body surface.

The transport epithelium that enables the albatross and other marine birds to survive on seawater remained undiscovered for many years. To find it, researchers gave captive marine birds only seawater to drink. Although very little salt appeared in the birds' urine, tests revealed that fluid dripping from the tip of their beaks was a concentrated solution of salt (NaCl). The source of this solution turned out to be a pair of nasal salt glands packed with transport epithelia **(Figure 44.5)**. Salt glands, which are also found in sea turtles and marine iguanas, use active transport of ions to secrete a fluid much saltier than the ocean. Even though drinking seawater brings in a lot of salt, the salt gland enables these marine vertebrates to achieve a net gain of water. By contrast, humans who drink a given volume of seawater must use a *greater* volume of water to excrete the salt load, with the result that they become dehydrated.

Figure 44.5

Salt secretion in the nasal glands of a marine bird.

In the nasal salt gland, salt moves from the blood in capillaries through the secretory cells of the transport epithelium and into the lumen of secretory tubules and is secreted.

A transport epithelium moves salt from the blood into secretory tubules, which drain into central ducts leading to the nostrils.

Exploded view of the nasal salt gland from the nostril (with salt secretions) near the base of the beak of an albatross. Blood vessels surround the secretory tubules inside the transport epithelium. As blood flows through the capillaries, salt from the blood travels across the secretory cells of the transport epithelium and into the lumen of the secretory tubule. Salt then drains into the central ducts leading to the nostrils and is secreted.

Transport epithelia that function in maintaining water balance also often function in disposal of metabolic wastes. We'll see examples of this coordinated function in our upcoming consideration of earthworm and insect excretory systems as well as the vertebrate kidney.